

**Problem 5**

Let  $x, h \in N_G(H)$

Consider  $xhx^{-1} \in N_G(H)$  since  $N_G(H)$  is a subgroup thus  $x \in N_G(H)$  thus  $N_G(H) \subseteq N_G(N_G(H))$

We know  $H \trianglelefteq N_G(H) \trianglelefteq N_G(N_G(H))$  thus by proven in class since  $H$  is a  $p$ -Sylow subgroup of  $N_G(H)$  we know  $H \trianglelefteq N_G(N_G(H))$

Thus let  $x \in N_G(N_G(H))$

Consider  $xHx^{-1} = H$  thus  $x \in N_G(H)$  thus  $N_G(N_G(H)) \subseteq N_G(H)$

Thus  $N_G(N_G(H)) = N_G(H)$

## Problem 6

Let  $|G| = pqr$  where  $p, q, r$  are distinct primes.

WLOG: Let  $p < q < r$  and for contradiction suppose  $G$  has no normal Sylow subgroups thus  $n_r, n_q, n_p > 1$

By Sylow 3 we know  $n_r \mid pq$  and  $n_r \equiv 1 \pmod{r}$

Thus,  $n_r = pq$  since  $n_r \equiv 1 \pmod{r}$  and  $p < q < r$

Thus, there are  $pq(r-1) = pqr - pq$  elements of order  $r$ .

By Sylow 3 we know  $n_q \mid pr$  and  $n_q \equiv 1 \pmod{q}$

Thus,  $n_q \geq r$  since  $n_q \equiv 1 \pmod{q}$  and  $p < q < r$

Thus, there are at least  $r(q-1) = qr - r$  elements of order  $q$ .

By Sylow 3 we know  $n_p \mid qr$  and  $n_p \equiv 1 \pmod{p}$

Thus,  $n_p \geq q$  since  $n_p \equiv 1 \pmod{p}$  and  $p < q < r$  Thus, there are at least  $q(p-1) = pq - q$  elements of order  $p$ .

Thus,  $pqr - pq + qr - r + pq - q \leq pqr$

Thus,  $qr - r - q \leq 0 \implies q(r-1) \leq r \implies q \leq \frac{r}{r-1}$

Since  $2 \leq p < q$  we know  $2 < q \leq \frac{r}{r-1} \leq 2$  since  $r > r-1$

This is a contradiction thus there is a normal Sylow subgroup.

Proven in class we know groups of order  $p$  are simple, so we can rule these out.

- 61,67,71,73,79,83,89,97,101,103,107,109,113,127,131,137,139,149,151,157,163,167

Proven in class we know groups of order  $p^n$  have a nontrivial center thus they are not simple, so we can rule these out.

- 64,81,121,125,128

Proven in class we know groups of order  $pq$  are not simple, so we can rule these out.

- 62,65,69,74,77,82,85,86,87,91,93,94,95,106,111,115,118,119,122,123,129,133,134,141,142,143,145,146,155,158,159,161,166

Proven above we know groups of order  $pqr$  are not simple, so we can rule these out.

- 66,70,78,102,105,110,114,130,138,154,165

Proven in class we know groups of order  $p^2q$  are not simple, so we can rule these out,

- 63,68,75,76,92,98,99,116,117,124,147,148,153,164

Thus we have 72,80,84,88,90,96,100,104,108,112,120,126,132,135,136,140,144,150,152,156,160,162 left

- 72: Let  $n_3$  be the number of 3-Sylow subgroups.  
By Sylow 3,  $n_3 = 1, 4$  if  $n_3 = 1$  then done so assume  $n_3 = 4$   
Thus  $G$  acts of these 4 subgroups by conjugation, so there is a homomorphism from  $G \rightarrow S_4$   
Since  $|G| = 72 > |S_4| = 24$  thus this map is not injective thus its kernel is not trivial, so that is a normal subgroup of  $G$ .
- 80: Let  $n_5$  be the number of 5-Sylow subgroups.  
By Sylow 3,  $n_5 = 1, 16$  if  $n_5 = 1$  then done so assume  $n_5 = 16$   
Thus there are  $16 * 4 = 64$  elements of order 5 and  $80 - 64 = 16$  elements left thus the remaining elements form the unique 2-Sylow subgroup thus  $G$  is not simple.
- 84: Let  $n_7$  be the number of 7-Sylow subgroups.  
By Sylow 3,  $n_7 = 1$  so it is normal thus done
- 88: Let  $n_{11}$  be the number of 11-Sylow subgroups.  
By Sylow 3,  $n_{11} = 1$  so it is normal thus done
- 90:
- 96: Let  $n_2$  be the number of 2-Sylow subgroups.  
By Sylow 3,  $n_2 = 1, 3$  if  $n_2 = 1$  then done so assume  $n_2 = 3$   
Thus  $G$  acts of these 3 subgroups by conjugation, so there is a homomorphism from  $G \rightarrow S_3$   
Since  $|G| = 96 > |S_3| = 6$  thus this map is not injective thus its kernel is not trivial, so that is a normal subgroup of  $G$ .
- 100: Let  $n_5$  be the number of 5-Sylow subgroups.  
By Sylow 3,  $n_5 = 1$  so it is normal thus done
- 104: Let  $n_{13}$  be the number of 13-Sylow subgroups.  
By Sylow 3,  $n_{13} = 1$  so it is normal thus done
- 108: Let  $n_3$  be the number of 3-Sylow subgroups.  
By Sylow 3,  $n_3 = 1, 4$  if  $n_3 = 1$  then so assume  $n_3 = 4$   
Thus  $G$  acts of these 4 subgroups by conjugation, so there is a homomorphism from  $G \rightarrow S_4$   
Since  $|G| = 108 > |S_4| = 24$  thus this map is not injective thus its kernel is not trivial, so that is a normal subgroup of  $G$ .
- 112:
- 120:
- 126: Let  $n_7$  be the number of 7-Sylow subgroups.  
By Sylow 3,  $n_7 = 1$  so it is normal thus done

- 132: Let  $n_{11}$  be the number of 11-Sylow subgroups.  
 By Sylow 3,  $n_{11} = 1, 12$  if  $n_{11} = 1$  then done so assume  $n_{11} = 12$   
 Thus there are  $12 * 10 = 120$  elements of order 11 and  $132 - 120 = 12$  elements left  
 Let  $n_3$  be the number of 3-Sylow subgroups.  
 By Sylow 3,  $n_3 = 1, 4, 22$  if  $n_3 = 1$  then done so assume  $n_3 \neq 1$   
 If  $n_3 = 4$  then there are  $4 * 2 = 8$  elements of order 3 thus there are  $12 - 8 = 4$  elements left  
 thus the remaining elements form the unique 2-Sylow subgroup.  
 If  $n_3 = 22$  then there are  $22 * 2 = 44$  elements of order 3 this is too many and a contradiction
- 135: Let  $n_5$  be the number of 5-Sylow subgroups.  
 By Sylow 3,  $n_5 = 1$  so it is normal thus done
- 136: Let  $n_{17}$  be the number of 17-Sylow subgroups.  
 By Sylow 3,  $n_{17} = 1$  so it is normal thus done
- 140: Let  $n_7$  be the number of 7-Sylow subgroups.  
 By Sylow 3,  $n_7 = 1$  so it is normal thus done
- 144: Let  $n_3$  be the number of 3-Sylow subgroups.  
 By Sylow 3,  $n_3 = 1, 4, 16$  if  $n_3 = 1$  then so assume  $n_3 \neq 1$   
 Thus  $G$  acts on these 4 subgroups by conjugation, so there is a homomorphism from  $G \rightarrow S_4$   
 Since  $|G| = 144 > |S_4| = 24$  thus this map is not injective thus its kernel is not trivial, so that  
 is a normal subgroup of  $G$ .
- 150: Let  $n_5$  be the number of 5-Sylow subgroups.  
 By Sylow 3,  $n_5 = 1, 6$  if  $n_5 = 1$  then done so assume  $n_5 = 6$   
 Thus  $G$  acts on these 6 subgroups of conjugation, so there is a homomorphism from  $G \rightarrow S_6$   
 If the homomorphism is injective this implies  $|\varphi(G)| = |G|$  and since  $\varphi(G)$  is a subgroup of  
 $S_6$  by lagrange's theorem its order must divide  $|S_6| = 720$   
 However,  $|G| = 150 \nmid 720$  thus this homomorphism cannot be injective. If kernel is the  
 whole group this would mean there are 6 distinct normal 5-Sylow subgroups which is a  
 contradiction thus the kernel of this homomorphism is normal in  $G$
- 152: Let  $n_{19}$  be the number of 19-Sylow subgroups.  
 By Sylow 3,  $n_{19} = 1$  so it is normal thus done
- 156: Let  $n_{13}$  be the number of 13-Sylow subgroups.  
 By Sylow 3,  $n_{13} = 1$  so it is normal thus done

- 160: Let  $n_2$  be the number of 2-Sylow subgroups.

By Sylow 3,  $n_2 = 1, 5$  if  $n_2 = 1$  then done so assume  $n_2 = 5$

Thus  $G$  acts on these 5 subgroups by conjugation, so there is a homomorphism from  $G \rightarrow S_5$

Since  $|G| = 160 > |S_5| = 120$  thus this map is not injective thus its kernel is not trivial, so that is a normal subgroup of  $G$ .

- 162: Let  $n_3$  be the number of 3-Sylow subgroups.

By Sylow 3,  $n_3 = 1$  so it is normal thus done